



## Solve with us 6.1 - Not Graded



This assignment will not be graded and is only for practice.

Key Points: 1

- A map  $T: V \rightarrow W$  between two vector spaces  $V$  and  $W$  is said to be a linear transformation if for any two vectors  $v_1$  and  $v_2$  in the vector space  $V$  and for any scalar  $c$  in  $R$  the following conditions hold:
  - $T(v_1 + v_2) = T(v_1) + T(v_2)$
  - $T(cv_1) = cT(v_1)$

1 point

Let  $T$  be a function from  $R^2$  to  $R^2$  defined as follows:

$$T: R^2 \rightarrow R^2 \quad T(x, y) = (2x, y)$$

Choose the set of correct option(s).

[Hint: Assume  $v_1 = (x_1, y_1)$  and  $v_2 = (x_2, y_2)$ , then check the properties of linear transformation for the given function  $T$ .]

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$T(v_1 + v_2) = T(v_1) + T(v_2)$ , for all  $v_1, v_2 \in R^2$ .

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$T(cv_1) = cT(v_1)$  for all  $v_1 \in R^2$ , and  $c \in R$ .

☐

$T$  is a linear transformation.

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$T$  is not a linear transformation.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T(v_1 + v_2) = T(v_1) + T(v_2)$ , for all  $v_1, v_2 \in R^2$ .

$T(cv_1) = cT(v_1)$  for all  $v_1 \in R^2$ , and  $c \in R$ .

$T$  is a linear transformation.

1 point

Let  $T$  be a function from  $R^2$  to  $R^2$  defined as follows:

$$T: R^2 \rightarrow R^2 \quad T(x, y) = (x - 1, y + 1)$$

Choose the set of correct option(s).

[Hint: Assume  $v_1 = (x_1, y_1)$  and  $v_2 = (x_2, y_2)$ , then check the properties of linear transformation for the given function  $T$ .]

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$T(v_1 + v_2) = T(v_1) + T(v_2)$ , for all  $v_1, v_2 \in R^2$ .

☐

$T(cv_1) = cT(v_1)$  for all  $v_1 \in R^2$ , and  $c \in R$ .

☐

$T$  is a linear transformation.

☐

$T$  is not a linear transformation.



**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$  is not a linear transformation.

Key Points: 2

- A linear transformation  $T: V \rightarrow W$  is said to be an isomorphism if it is bijective, i.e. the followings hold:
  - $T(v_1) = T(v_2)$  implies  $v_1 = v_2$  (Injectivity).
  - For any  $w \in W$  there exists  $v \in V$  such that  $T(v) = w$  (Surjectivity).

1 point

If  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x, 0)$ , then which of the following options is true?

[[Hints:

Hint 1: Let  $v_1 = (x_1, y_1)$ ,  $v_2 = (x_2, y_2)$ , and assume that  $T(v_1) = T(v_2)$ . Then check whether  $v_1 = v_2$  or not.

Hint 2: Does there exist any  $v \in \mathbb{R}^2$  for which  $T(v) = (0, 1)$ ?

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$T$  is both one to one and onto.

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$T$  is one to one but not onto.

☐

$T$  is onto but not one to one.

☐

$T$  is neither one to one nor onto.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$  is neither one to one nor onto.

1 point

If  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$  is a linear transformation defined by

$T(x, y) = (2x + 3y, 5x - y, x + 6y)$ . Which of the following options is true?

[Hint: Does there exist any  $v \in \mathbb{R}^2$  for which  $T(v) = (1, 0, 0)$ ?

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$T$  is both one to one and onto.

☐

$T$  is one to one but not onto.

☐

$T$  is onto but not one to one.

☐

$T$  is neither one to one nor onto.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$  is one to one but not onto.

Key Points: 3Key Points: 3

- Standard Ordered basis of  $R^n$   $R^n$ :

$$\{e_i \mid i = 1, 2, \dots, n\}, \{e_i \mid i = 1, 2, \dots, n\},$$

where  $e_1 = (1, 0, 0, \dots, 0)$ ,  $e_2 = (0, 1, 0, \dots, 0)$ ,  $e_3 = (0, 0, 1, \dots, 0)$ ,  $\dots$ ,  $e_n = (0, 0, \dots, 0, 1)$ . In general,  $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)$  where 1 is at the  $i$ -th position and 0 elsewhere.

- If  $T(e_i)$  is known for all  $i$ , then the description of  $T$  can be known entirely. Suppose  $T(e_i) = v_i$ , for  $i = 1, 2, \dots, n$ , then we have

$$\begin{aligned} T(x_1, x_2, \dots, x_n) &= T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) = T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) \\ &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) = x_1 v_1 + x_2 v_2 + \dots + x_n v_n \\ &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) = x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) = x_1 v_1 + x_2 v_2 + \dots + x_n v_n \end{aligned}$$

1 point

Let  $T: R^2 \rightarrow R^2$  be a linear transformation, such that  $T(1, 0) = (3, 2)$  and  $T(0, 1) = (1, 5)$ . Which of the following options is true?

[Hint:  $\{(1, 0), (0, 1)\}$  is the standard ordered basis of  $R^2$ .]

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$$T(x, y) = (3x + y, 2x + 5y)$$

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$$T(x, y) = (3x + 2y, x + 5y)$$

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$$T(x, y) = (5x, 6y)$$

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$$T(x, y) = (4x, 7y)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(x, y) = (3x + y, 2x + 5y)$$

1 point

Let  $T: R^2 \rightarrow R^2$  be a linear transformation, such that  $T(2, 1) = (3, 2)$  and  $T(1, 3) = (0, 0)$ . Which of the following options is true?

[Hint: Express  $(1, 0)$  and  $(0, 1)$  in terms of  $(2, 1)$  and  $(1, 3)$ .]

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$$T(x, y) = \frac{1}{5}(3x, 2y)$$

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$$T(x, y) = (x + 1, y + 1)$$

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$$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$$

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$T$  cannot be determined from the given information.

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

$$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y) \quad T(x, y) = 51(9x - 3y, 6x - 2y)$$

Key Points: 4

- Let  $T: V \rightarrow W$  be a linear transformation. Let  $\beta = \{v_1, v_2, \dots, v_n\}$  be an ordered basis of  $V$  and  $\gamma = \{w_1, w_2, \dots, w_m\}$  be an ordered basis of  $W$ . So each  $f(v_i)$  can be uniquely written as linear combination of  $w_j$ 's, where  $i = 1, 2, \dots, n$  and  $j = 1, 2, \dots, m$ .

$$\begin{aligned} T(v_1) &= a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m & T(v_2) &= a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m & \vdots & T(v_n) = a_{1n}w_1 \\ T(v_1) &= a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m & T(v_2) &= a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m & \vdots & T(v_n) = a_{1n}w_1 \\ &+ a_{2n}w_2 + \dots + a_{mn}w_m \end{aligned}$$

The matrix corresponding to the linear transformation  $T$  with respect to the ordered bases  $\beta$  and  $\gamma$  is given by,

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

The above matrix not only depends on the linear map  $T$ , but also on the choice of the ordered bases  $\beta$  for  $V$  and  $\gamma$  for  $W$ .

- It is important to note that, for a fixed choice of basis of  $V$  and  $W$ , there is a one to one correspondence between the set of all linear transformations from  $V$  to  $W$ , and the set of all  $m \times n$  matrices, where  $n = \dim(V)$  and  $m = \dim(W)$ .

1 point

Suppose  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x, 0)$ . Which of the following matrices corresponds to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain?

[Hint:  $T(1, 0) = (1, 0)$  and  $T(0, 1) = (0, 0)$ ]

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$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

1 point

Suppose  $T: R^2 \rightarrow R^2$  is a linear transformation, such that  $T(x, y) = (x + y, x - y)$ . Which of the following matrices corresponds to  $T$  with respect to the standard ordered basis of  $R^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for the domain and  $\{(1, 1), (1, -1)\}$  for the co-domain?

[Hint:  $T(1, 0) = (1, 1)$  and  $T(0, 1) = (1, -1)$ ]

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$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

☐

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

•  $T(1, 0) = (1, 1) = 1(1, 1) + 0(1, -1)$ , and  $T(0, 1) = (1, -1) = 0(1, 1) + 1(1, -1)$ .

**Accepted Answers:**

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1 point

Let  $W = \{(x, y, z) \mid x = 2y + z\}$  be a subspace of  $R^3$ . Let  $\beta = \{(2, 1, 0), (1, 0, 1)\}$  be a basis of  $W$ . Let  $T: W \rightarrow R^2$  be a linear transformation such that  $T(2, 1, 0) = (1, 0)$  and  $T(1, 0, 1) = (0, 1)$ . What will be the matrix representation of  $T$  with respect to the basis  $\beta$  for  $W$  and  $\gamma = \{(1, 1), (1, -1)\}$  for  $R^2$ ?

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$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix}$$

1 point

Let  $\beta = \{(1, 0), (0, 1)\}$  and  $\gamma = \{(1, 1), (1, -1)\}$  be two bases of  $R^2$ . If  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  is the matrix representation of a linear transformation  $T: R^2 \rightarrow R^2$  with respect to  $\beta$  for the both the domain and co-domain. What will be the matrix representation of  $T$  with respect to  $\gamma$  for the domain and  $\beta$  for the co-domain?

[Hint: First find  $T(x, y)$  and then find  $T(1, 1)$  and  $T(1, -1)$ .]

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$$\begin{bmatrix} a + b & c + d \\ a - b & c - d \end{bmatrix}$$

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$$\begin{bmatrix} a - b \\ c - d \end{bmatrix}$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

•  $T(x, y) = (ax+by, cx+dy)$ , so  $T(1, 1) = (a+b, c+d)$  and  $T(1, -1) = (a-b, c-d)$ .

**Accepted Answers:**

$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix}$$